M.Tech End Term Practical Examination

Subject- Machine Learning Semester-I

3. Using the UCI Automobile dataset, build and train an Artificial Neural Network (ANN) to predict the price of a car based on its features.

Tasks:

i. Split the dataset into training and testing sets .
ii. Train your model and evaluate performance using MSE and R^2 score.
iii. Plot predicted vs. actual prices for the test set.

CSCN1103

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Test-1

Duration: 45 Min. **Marks: 10**
Subject: MACHINE LEARNING
Code: AI-103 **Subject Coordinator:** Prof. R.K. Pateriya
Course: M.Tech 1st Year (1st Sem) **Session:** Aug-Dec 2025-2026 **Date:** 26/09/2025
Name: **Scholar No.:** **Signature:**

Important Instructions:
1. All questions are compulsory.
2. Each answer should include technical justification and relevant concepts.
3. Your test may be canceled if any unfair means are detected during the test.

1. Explain the concept of multiple regression. Given the following dataset, determine the coefficient matrix (β) of the multiple regression model using the ordinary least squares method. (2.5 Marks)

Y	100	108	120	131	145	155	160
X_1	45	67	85	23	54	43	23
X_2	25	81	14	15	16	11	64

2. Calculate silhouette scores using Euclidian distance for the following scenarios and interpret the results:
$S = \frac{b_i - a_i}{\max(b_i, a_i)}$
Scenario 1: Four points: A(1, 1), B(1.5, 1), and C(1, 1.5) are in Cluster 1, and D(10, 10) is in Cluster 2. Calculate the silhouette score for point A. How does the distant outlier affect the score?
Scenario 2:
Five points: D1(1, 2), D2(1.5, 2.5), and D3(2, 3) are in Cluster 1, and D4(5, 5) and D5(6, 6) are in Cluster 2. Calculate the average silhouette score for Cluster 1. (2.5 Marks)

3. Find the similarity between documents 1 and 2 by Simple matching coefficient and Jaccard coefficient.
D 1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)
D 2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)
Consider following dataset

Object	Color	Distance
1	R	L
2	B	M
3	G	M
4	R	L

Obtain dissimilarity matrix considering both the categorical attributes, color and position.(2.5 Marks)

4. Consider k-means algorithm with k=2 is applied to following 2D-dataset Y which contains outliers Y={(-100,-100), (0,0), (1,1), (0, 1), (1, 2), (5, 4), (5,5), (5,6)}.
How do outliers (e.g. the point (-100,-100)) impact the k-means clustering result? (2.5 Marks)

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Mid Sem Examination **MONTH and YEAR: Oct-Nov 2025**

Course: M. Tech. **Semester:** I **Branch:** AC/CN/IS/AI
Subject Code: CAI AI 1102 / CS IS 1152 / CS CN 1152 / CS AC 1152
Subject Name: Machine Learning and Deep Learning **Maximum Marks:** 20
Duration: 1 hour **Date: 30/10/2025** **Time: 9:30 to 10:30 A.M.**

Note: *Answer all questions. Make necessary assumptions wherever required.*

Q. No.	Questions	Marks	COs
1.	A researcher wants to predict the house price (in lakhs) based on two factors - size of the house (in sq. ft) and number of bedrooms. The following data were collected for five houses:	6	CO1

House	Size (X_1)	Bedrooms (X_2)	Price (Y)
1	1400	3	245
2	1600	3	312
3	1700	4	279
4	1875	4	308
5	1100	2	199

(a) Compute the coefficient matrix (β) of the multiple regression model.
(b) Predict the house price for a house of size 1800 sq. ft with 3 bedrooms.
(c) Compute R^2 and comment on the model's performance.

2.	Five points are given: A(1,1), B(1,3), C(5,3), D(6,2), E(6,5). Perform Agglomerative Clustering using Average Linkage as follows: 1. Compute the distance matrix. 2. Show merging sequence and cluster distances. 3. Derive the final clusters when the cut height = 3. 4. Comment on which linkage (single, complete, or average) would better preserve cluster compactness in this dataset and why?	04	CO1
3.	Consider three datasets such as all datapoints are forming: D1: One compact cluster D2: Two well-separated clusters D3: One large cluster and one small cluster Explain and justify which method (K-Means vs. K-Means++) will perform better for each dataset, using the concept of initial centroid selection probability and expected SSE.	05	CO2

4.	Given a bivariate dataset: (25,2), (45,20), (20,10), (30,15), (50,25) Compute the Mahalanobis distance of point (30, 15) from the data distribution. Show all steps including: 1. Mean vector 2. Covariance matrix 3. Inverse of covariance matrix 4. Mahalanobis distance formula $D_M(x) = \sqrt{(x-\mu)^T S^{-1}(x-\mu)}$	05	CO1

**********END**********

$D_m(x) = \frac{|x-\mu|}{\sigma}$

$X = \begin{bmatrix} 25,2 \\ 45,20 \\ 20,10 \\ 30,15 \\ 50,25 \end{bmatrix}$ $\leftarrow (30, 15)$

$S =$

x	y	$x-\bar{x}$	$y-\bar{y}$	$x-\bar{x} \times y-\bar{y}$
25	2	-9	-12.4	111.6
45	20	11	5.6	61.6
20	10	-14	-4.4	61.6
30	15	-4	0.6	-2.4
50	25	16	10.6	169.6

$x = (30, 15)$

$S^2 = \frac{\Sigma(x_i - \bar{x})^2}{N-1}$

(i)

$\mu_1 = 34$, $\mu_2 = 14.4$ — mean vector $= [34, 14.4]$

$\mu_2 = 14.4$

$\text{cov}(1,2) = \frac{\Sigma(x_i-\bar{x})(y_i-\bar{y})}{n-1}$

- covariance matrix $= \begin{bmatrix} \text{var}(1) & \text{cov}(1,2) \\ \text{cov}(1,2) & \text{var}(2) \end{bmatrix}$

(ii)

$x - \mu =$

$[30-34, 15-14.4]$

$\begin{bmatrix} -4 & 0.6 \end{bmatrix}_{2,1}$

$\begin{bmatrix} -4 \\ 0.6 \end{bmatrix} - A$

$S^{-1} = \frac{1}{|S|} \text{adj}(S)$

$S = \begin{bmatrix} 167.5 & 100.5 \\ 100.5 & 79.3 \end{bmatrix}$

$= \frac{1}{3192.5} \begin{bmatrix} 79.3 & -100.5 \\ -100.5 & 167.5 \end{bmatrix}$

$= \frac{1}{3192.5} \begin{bmatrix} 79.3 & -100.5 \\ -100.5 & 167.5 \end{bmatrix}_{2/1} = S^{-1} \begin{bmatrix} 0.0249 & -0.0315 \\ -0.0315 & 0.052 \end{bmatrix}_{2\times2}$

$A \times S^{-1} = \begin{bmatrix} -0.012 \\ 0.1552 \end{bmatrix}_{2\times1} \times [-4, 0.6]_{1\times2}$

Name of the Student..... Scholar Number..2S21201110

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: END TERM **December 2025**

Course: M.Tech **Semester: I** **Subject Code: CS AC/CN/IS 1152, CAI AI 1102**
Subject Name: Machine Learning and Deep Learning **Maximum Marks: 40**
Duration: 2Hrs **Date: 08/12/25** **Time: 9:30 AM to 11:30 AM**

Note: *All questions are compulsory. Make necessary assumptions wherever required.*

Q. No.	Questions	Marks	CO
1.(A)	Design a Naïve Bayes classifier for the following data. Find the prediction for the instance: Income = Medium, Credit Score = Poor, Age Group = Adult.	5	2

Income	High	Low	Medium	High	Medium	Low
Credit Score	Good	Poor	Good	Poor	Good	Poor
Age group	Adult	Young	Senior	Adult	Young	Senior
Approve	Yes	No	Yes	No	Yes	Yes

(B)	The table below lists a dataset that contains physical and categorical properties of fruits. Assuming that the model uses Euclidean distance to find nearest neighbor (k=3), what prediction will the model return for the query instances: Weight = 135 g, Sweetness = 6, Color = Red?	5	2

Weight(g)	120	150	130	160	110	140
Sweetness	8	5	7	6	9	4
Color	Red	Green	Red	Yellow	Red	Green
Fruit Type	Apple	Pear	Apple	Mango	Apple	Pear

2.(A)	Design a decision tree classifier using Gini Index that predicts whether a patient has a severe illness using the following training dataset:	5	2

Fever	High	High	Mild	High	Mild	Mild	High	Mild
Cough	Yes	No	Yes	Yes	No	Yes	No	No
Test Result	+	+	-	-	+	+	-	-
Severe	Yes	Yes	No	No	No	Yes	No	No

(B)	What are the different model-combination strategies used in ensemble learning? Describe each method and explain how they contribute to improving predictive performance. Which method supports parallelism during the classification process?	5	2

3.(A)	Explain the roles of loss function, epochs, and learning rate in training an Artificial Neural Network (ANN). Describe how the loss function measures model performance, what happens during multiple epochs, and how learning rate controls weight updates during backpropagation. Discuss why a high learning rate might cause training to diverge, while a low one leads to slow convergence?	5	1
(B)	A neural network binary classifier uses two input features (x_1, x_2) and one output neuron with a sigmoid activation. For each epoch you are given the neuron's weights (w_1, w_2) and bias. For each epoch: i. Compute the neuron's output for each sample using the given weights and bias, ii. Apply the sigmoid activation to obtain the predicted probability $\hat{y}$ for each sample, iii. Calculate the Log Loss for that epoch using the 4 samples, and iv. Finally, report the average (mean) Log Loss across the two epochs. **Weights for each epoch:** **Epoch 1:** $w_1 = 0.50$, $w_2 = 0.40$, bias = 0.10 **Epoch 2:** $w_1 = 0.60$, $w_2 = 0.45$, bias = 0.15 Dataset:	5	1

Sample	x_1	x_2	True Label (y)
1	0.6	0.2	1
2	0.3	0.9	0
3	0.8	0.4	1
4	0.1	0.5	0

4.(A)	A researcher collected data on the number of hours studied (X) and the exam scores (Y)	5	1

Student	Hours Studied (X)	Exam Score (Y)
1	10	70
2	12	80
3	8	60
4	14	90

Calculate the Pearson correlation coefficient (r) between X and Y using the raw score formula. Interpret the strength and direction of the relationship. If the hours studied (X) were all halved and the exam scores (Y) were all increased by 10 points, how would the new Pearson correlation coefficient compare to the original one? Justify your answer theoretically without recalculating.

(B)	Consider a small 1D dataset with three points: $P_1(2)$, $P_2(5)$, $P_3(8)$. Assume the data is clustered into k=2 clusters: $C_1 = \{P_1, P_2\}$ and $C_2 = \{P_3\}$. The centroids are $M_1=3.5$ and $M_2=8$. Explain why the calculated average score and the individual score for P_2 suggest that $k=2$ might not be the optimal choice for this data distribution.	5	1

**********END**********